THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL UNIT TEST SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Sound needs a: 1

(a) light **(b)** vacuum **(c)** medium **(d)** magnet

2. The slope of a v-t graph gives: 1

(a) distance **(b)** force **(c)** acceleration **(d)** speed

3. A rocket works on the: 1

(a) second law **(b)** law of gravitation only

(c) first law **(d)** third law

4. The fulcrum is the: 1

(a) pivot of a lever **(b)** effort **(c)** force **(d)** load

5. Infrasonic sounds have frequency: 1

(a) 100 Hz **(b)** below 20 Hz **(c)** 20 kHz **(d)** above 20 kHz

6. A man walks 10 m in 5 s. His speed is: 1

(a) 0.5 m/s **(b)** 50 m/s **(c)** 2 m/s **(d)** 5 m/s

7. A mass of 2 kg accelerates at 3 m/s squared. Force: 1

(a) 5 N **(b)** 6 N **(c)** 1.5 N **(d)** 0.67 N

8. Work done in 10 s is 200 J. Power: 1

(a) 2000 W **(b)** 0.05 W **(c)** 210 W **(d)** 20 W

9. Time period of 10 Hz is: 1

(a) 100 s **(b)** 0.1 s **(c)** 10 s **(d)** 1 s

10. Speed = 1

(a) time/distance **(b)** mass/volume **(c)** distance x time **(d)** distance/time

11. The first law is also called the law of: 1

(a) inertia **(b)** gravitation **(c)** momentum **(d)** energy

12. Unit of power: 1

(a) newton **(b)** watt **(c)** joule **(d)** pascal

13. Sound cannot travel in: 1

(a) water **(b)** air **(c)** steel **(d)** vacuum

14. A runner completes one lap of a circular track. The displacement is: 1

(a) half **(b)** zero
(c) double **(d)** equal to distance

15. A body at rest remains at rest due to: 1

(a) friction **(b)** inertia **(c)** gravity **(d)** force

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is momentum? 2

17. State the law of conservation of energy. 2

18. What is wavelength? 2

19. Write the second equation of motion. 2

20. Differentiate between mass and weight in two points. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the time period of 250 Hz. 3

22. Name three scalars and three vectors. 3

23. Find the net force when 10 N east and 4 N west act on a body. 3

24. State the energy change in a falling stone. 3

OR

State the effect of temperature on the speed of sound. 3

25. List three differences between speed and velocity. 3

OR

Give three examples of inertia of motion. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the average speed: 60 km in 2 h and 40 km in 1 h. 5

OR

Explain the working of a rocket. 5

27. Explain the inclined plane. 5

OR

Explain the audible range and the infrasonic and ultrasonic sounds. 5

END OF QUESTION PAPER Total: 50 Marks

INTERMIND